Week 3: Introduction to Data Analysis- Working with Real Data

LOAD DATA

```python
import pandas as pd

# ------------------------------
# STEP 1: LOAD DATA
# ------------------------------
try:
    df = pd.read_csv('sales_data.csv')
    print(" Data loaded successfully!\n")
except FileNotFoundError:
    print(" Error: sales_data.csv file not found!")
    exit()

'''
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code

```
PS C:\Users\kaushal kumar\OneDrive\Desktop\Week_3> python -u "c:\Users\kaushal kumar\OneDrive\Desktop\Week_3\sales_analysis.py"
 Data loaded successfully!

PS C:\Users\kaushal kumar\OneDrive\Desktop\Week_3>
```

EXPLORE DATA

```python
# ------------------------------
# STEP 2: EXPLORE DATA
# ------------------------------
print("First 5 Rows:")
print(df.head(), "\n")

print("Shape of Data (Rows, Columns):")
print(df.shape, "\n")

print("Column Names:")
print(df.columns, "\n")

print("Data Info:")
print(df.info(), "\n")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code

```
PS C:\Users\kaushal kumar\OneDrive\Desktop\Week_3> python -u "c:\Users\kaushal kumar\OneDrive\Desktop\Week_3\sales_analysis.py"
Data Info:
<class 'pandas.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 7 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Date         100 non-null    str
 1   Product      100 non-null    str
 2   Quantity     100 non-null    int64
 3   Price        100 non-null    int64
 4   Customer_ID  100 non-null    str
 5   Region       100 non-null    str
 6   Total_Sales  100 non-null    int64
dtypes: int64(3), str(4)
memory usage: 8.3 KB
None
```

DATA CLEANING

```python
# ------------------------------
# STEP 3: DATA CLEANING
# ------------------------------
print("Checking Missing Values:")
print(df.isnull().sum(), "\n")

# Fill missing values with 0 (you can change logic if needed)
df.fillna(0, inplace=True)

# Remove duplicate rows
df.drop_duplicates(inplace=True)

print("Data cleaning completed!\n")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code

```
PS C:\Users\kaushal kumar\OneDrive\Desktop\Week_3> python -u "c:\Users\kaushal kumar\OneDrive\Desktop\Week_3\sales_analysis
 6   Total_Sales  100 non-null    int64
dtypes: int64(3), str(4)
memory usage: 8.3 KB
None

Checking Missing Values:
Date           0
Product        0
Quantity       0
Price          0
Customer_ID    0
Region         0
Total_Sales    0
dtype: int64

Data cleaning completed!
```

REPORT

sales_analysis.py analysis_report.md

analysis_report.md > # Sales Data Analysis Report > ## Total Revenue

```md
# Sales Data Analysis Report

## Total Revenue
₹ 12,365,048.00

## Average Sales
₹ 123,650.48

## Highest Sale
₹ 373,932.00

## Lowest Sale
₹ 6,540.00

## Best Selling Product
Laptop

## Insights
- This report shows overall sales performance.
- Best product contributes highest revenue.
- Data cleaned and processed using pandas.
```